

BSDS (2025 Semester II): Statistics II

Quiz 2

NAME :

Solution Set

ROLL :

Time: 75 minutes

Total marks: 30

1. Let $Z, X_1, \dots, X_n$ be i.i.d. samples from `normal(0, 1)` distribution. Consider the random variables

$$S_n = \frac{X_1^2 + \dots + X_n^2}{n}, \quad \text{and} \quad T_n = \frac{Z}{\sqrt{S_n}}.$$

- (a) What is the distribution of T_n ? Justify your answer.

[4]

As $x_i \stackrel{iid}{\sim} N(0, 1)$, $i=1, \dots, n$; $nS_n = \sum_{i=1}^n x_i^2 \sim \chi_n^2$.

Further, Z and $\{x_1, \dots, x_n\}$ are independent,
and hence Z and nS_n are independent.

Thus, $T_n = \frac{Z}{\sqrt{nS_n/n}} \sim t_n$ distribution.

(b) Identify the constant c , such that $S_n \xrightarrow{p} c$ as $n \rightarrow \infty$. Justify your answer.

[4]

By WLLN,

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \xrightarrow{p} E(x_i^2)$$

Now, $E(x_i^2) = \text{var}(x_i) = 1.$

Thus, $c = 1$

(c) Hence or otherwise show, with proper justification, that $T_n \xrightarrow{d} W$ as $n \rightarrow \infty$, where $W \sim \text{normal}(0, 1)$. [4]

We know, $Z \sim N(0, 1)$ and $S_n \xrightarrow{P} 1$,

by continuous mapping theorem $1/\sqrt{S_n} \xrightarrow{P} 1$.

Thus, by Slutsky's lemma

$$T_n = \frac{Z}{\sqrt{S_n}} \xrightarrow{d} W, \text{ where } W \sim N(0, 1).$$

2. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{normal}(\mu, \sigma^2)$, where σ^2 is ~~unknown~~. Consider the problem of testing $H_0 : \mu = 0$ against $H_1 : \mu > 0$.

Consider the test function of the form

$$\phi^*(x) = \begin{cases} 1 & \text{if } t_n \geq c_\alpha, \\ 0 & \text{otherwise,} \end{cases}$$

where t_n is a realization of the statistic

$$T_n = \frac{\sqrt{n}\bar{X}_n}{S_n}, \quad \text{and} \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

- (a) Find appropriate scaling of $\bar{X}_n$ and S_n^2 under which the distributions of the scaled random variables are $\text{normal}(0, 1)$ and χ_r^2 , respectively. Also, indicate the appropriate value of the degrees of freedom 'r'.
under H_0 . [1.5 + 2.5]

$$\bar{X}_n \sim N\left(0, \frac{\sigma^2}{n}\right)$$

$$\frac{\sqrt{n} \bar{X}_n}{\sigma} \sim N(0, 1)$$

Again,
$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi_{n-1}^2$$

So, the scaling factors are $\sigma/\sqrt{n}$ and $\sigma^2/(n-1)$, respectively,

and the d.f. $r = n-1$.

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(b) Using part (a) or otherwise find the exact distribution of T_n under H_0 . [3]

It is known that, $Z_n = \frac{\sqrt{n} \bar{X}_n}{\sigma}$ is independent of $W_n = \frac{(n-1)S_n^2}{\sigma^2}$. As $Z_n \sim N(0,1)$ and $W_n \sim \chi^2_{(n-1)}$,

$$T_n = \frac{Z_n}{\sqrt{\frac{W_n}{(n-1)}}} = \frac{\sqrt{n} \bar{X}_n}{S_n} \sim t_{n-1} \text{ distn.}$$

- (c) Find the size of the test ϕ^* . Hence, obtain the appropriate choice of c_α such that the size of ϕ^* is equal to α . [5]

$$\begin{aligned}\text{Size of } \phi^* &= E_{\mu=0}(\phi^*(z)) \\ &= P_{\mu=0}(T_n \geq c_\alpha) \\ &= 1 - F_{t;n-1}(c_\alpha),\end{aligned}$$

where $F_{t;n-1}$ is the CDF of t_{n-1} distribtn.

If ϕ^* is size- α then

$$\text{Size of } \phi^* = 1 - F_{t;n-1}(c_\alpha) = \alpha$$

$\therefore c_\alpha = t_{\alpha;n-1}$: the upper- α point of t_{n-1} distribtn.

(d) Based on Q.1, can you find the asymptotic distribution of T_n under H_0 ? [3]

From Q1, we get if $T_n^* \sim t_n$ distribution, then
as $n \rightarrow \infty$ $T_n^* \xrightarrow{d} W$ where $W \sim N(0,1)$ distr..

Here $T_n \sim t_{n-1}$ distribtn.

As $n-1 \rightarrow \infty$ whenever $n \rightarrow \infty$, $T_n \xrightarrow{d} N(0,1)$ distr..

- (e) How would the cutoff c_α change if the asymptotic distribution of T_n is used instead of the exact distribution? [3]

$$As \quad T_n \xrightarrow{d} N(0,1) \quad \text{under } H_0.$$

$$\text{Size of } \phi^* = P_{n=0} (T_n \geq c_\alpha)$$

$$\rightarrow 1 - \Phi(c_\alpha)$$

To make ϕ^* a size- α test, we choose c_α such that

$$1 - \Phi(c_\alpha) = \alpha$$

$\Rightarrow c_\alpha = z_\alpha$: the upper- α point of $N(0,1)$ distribution.

$$X \longrightarrow X$$

Rough work